

Last name _____

First name _____

LARSON—MATH 756—HOMEWORK WORKSHEET 05
Integer Programming Models for Independence Number

Let G be a graph with vertices $V(G) = \{v_1, \dots, v_n\}$. For each vertex v_i , associate a variable x_i . We want to maximize the sum of these variables subject to the constraints that, for each edge $v_i v_j$, the sum of the corresponding variables is no more than 1.

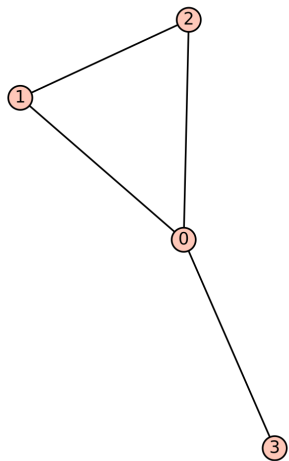
Vertex Packing Integer Program (VPIP)

$$\max \sum x_i$$

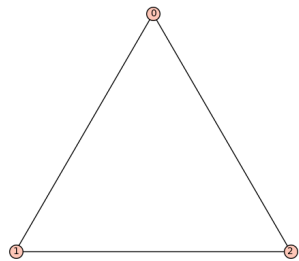
$$x_i + x_j \leq 1 \text{ (for } v_i v_j \in E(G)).$$

$$x_i \in \mathbb{Z}^{\geq 0}.$$

For **integer** programs we restrict the variables to be non-negative integers. Let ρ be the optimum value of this IP.



1. Write out the IP for the paw graph.
2. Why is the optimum IP value ρ bounded?
3. Why does the optimum IP value ρ exist?
4. Find the optimum IP value ρ for the paw graph. Argue that ρ is in fact optimum.
5. What is a *solution* (x_i values) that achieves this optimum.



6. Write out the IP for the complete graph K_3 .
7. Find the optimum IP value ρ for the K_3 and a solution that achieves this optimum. Argue that ρ is in fact optimum.
8. Prove: for any graph G the IP optimum ρ equals the independence number $\alpha(G)$.

We can *relax* an IP by allowing the variables to be real rather than integers. Let α_f be the LP optimum.

Vertex Packing Linear Program (VPLPP)

$$\max \sum x_i$$

$$x_i + x_j \leq 1 \text{ (for } v_i v_j \in E(G)\text{)}.$$

$$x_i \geq 0 \text{ (} x_i \in \mathbb{R}\text{)}.$$

9. Find the optimum LP value α_f for the paw graph. Argue that α_f is in fact optimum.
10. Find the optimum LP value α_f for the complete graph K_3 . Argue that α_f is in fact optimum.
11. How many edge constraints will the VPLP have for a complete graph K_n (with order n)?
12. Find the optimum LP value α_f for the complete graph K_n . Argue that α_f is in fact optimum.
13. Argue that, for any graph $\alpha \leq \alpha_f$.
14. Find and prove a tight upper bound for α_f .